IRT and Classical Item Analysis Report

Quiz 1 — summary for test authors

Based on the Stata log and exported diagnostic graphs

Executive summary

- The analysis used first attempts only: 59 students remained after removing 84 repeat attempts from the original 143 quiz records.
- Average performance was 3.68 out of 6 items correct. Most students scored between 3 and 5, so the quiz is not too difficult overall, but it has limited spread at the extremes.
- Internal consistency was low (Cronbach's alpha = 0.40). This is partly expected for a six-item quiz, but the item statistics suggest that some items may not be measuring the same target skill equally well.
- Item 1 was extremely easy (96.6% correct; Rasch difficulty = -4.42), contributing little to distinguishing among students except at the very low end.
- Items 5 and especially 6 had weak item-rest correlations (0.107 and 0.026). These should be reviewed first for ambiguity, scoring/key issues, or mismatch with the intended construct.
- The Rasch test information curve peaks around average ability, meaning the quiz measures middle-performing students better than very low- or very high-performing students.

Data and analysis approach

The dataset initially contained 143 quiz attempts across 41 variables. The analysis sorted records by student ID and attempt number, then kept only each student's first attempt. This left 59 independent student records for item analysis.

Each of the six scored questions was converted to a dichotomous item score: 1 = correct and 0 = incorrect. Items 1–4 were counted correct when the earned points equaled 15; items 5–6 were counted correct when the earned points equaled 20. The total score was the sum of the six binary item scores.

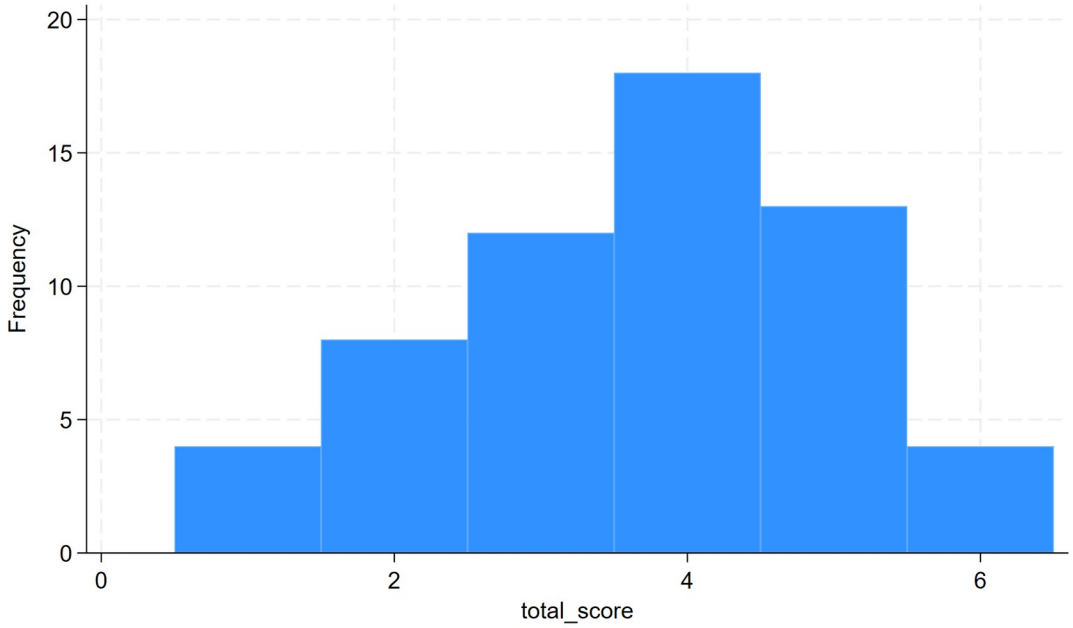
The report combines classical test statistics with a one-parameter logistic IRT model (Rasch / 1PL). In the Rasch model, all items share a common discrimination parameter and differ primarily in difficulty.

Overall test performance

Statistic	Value	Interpretation	Implication for authors
Number of students	59	Moderate sample for a classroom-level diagnostic analysis	Interpret results as practical evidence, not final calibration.
Mean total score	3.68 / 6	Average score ≈ 61.3%	The test was generally accessible to students.
Standard deviation	1.33	Moderate score spread	There is some differentiation, but a longer test would improve precision.
Observed range	1–6	No zero scores; some	The easiest item(s) may

		perfect scores	be too easy for this group.
Cronbach’s alpha	0.40	Low internal consistency	Review weak items and consider adding more well-targeted items.

Score distribution. The histogram shows a concentration around 4 correct answers, with relatively few very low and very high scores. The frequency table was: 1 = 4 students, 2 = 8, 3 = 12, 4 = 18, 5 = 13, 6 = 4.



Item-level findings

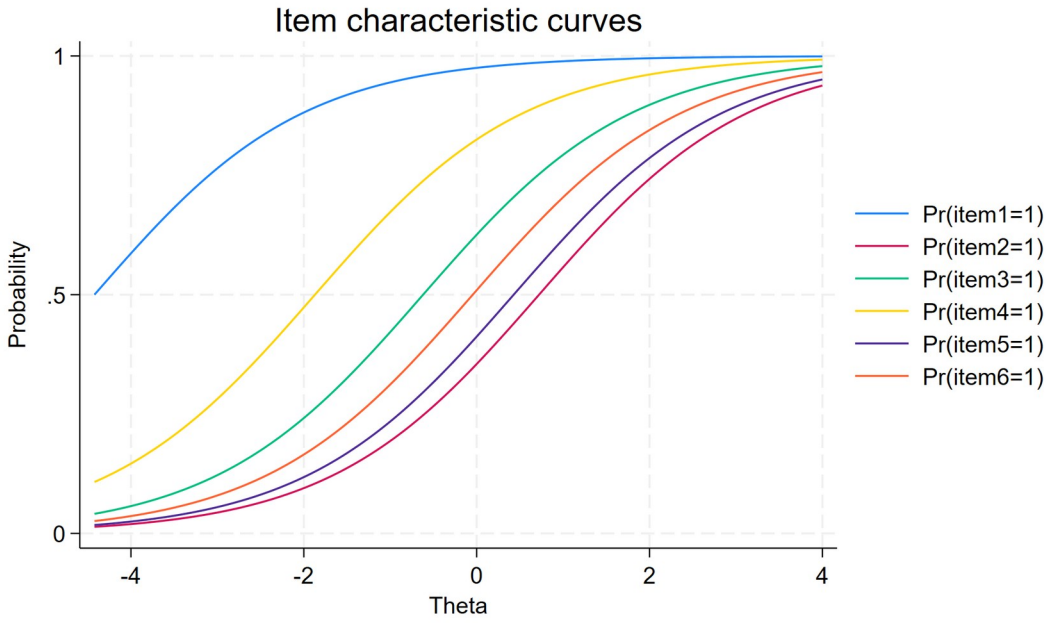
The table below combines the classical proportion correct, Rasch difficulty, item-rest correlation, and an author-facing recommendation. Negative Rasch difficulty values indicate easier items; positive values indicate harder items.

Item	Correct rate	Rasch difficulty	Item-rest corr.	Judgement	Recommended action
Item 1	96.6%	-4.42	0.254	Very easy	Keep only if intended as a warm-up / prerequisite check; otherwise increase challenge.
Item 2	37.3%	0.72	0.194	Hardest	Review wording, scoring key, and taught coverage; good candidate for revision if difficulty was unintended.
Item 3	61.0%	-0.62	0.306	Moderate-easy	Functioning reasonably; can remain with minor review.

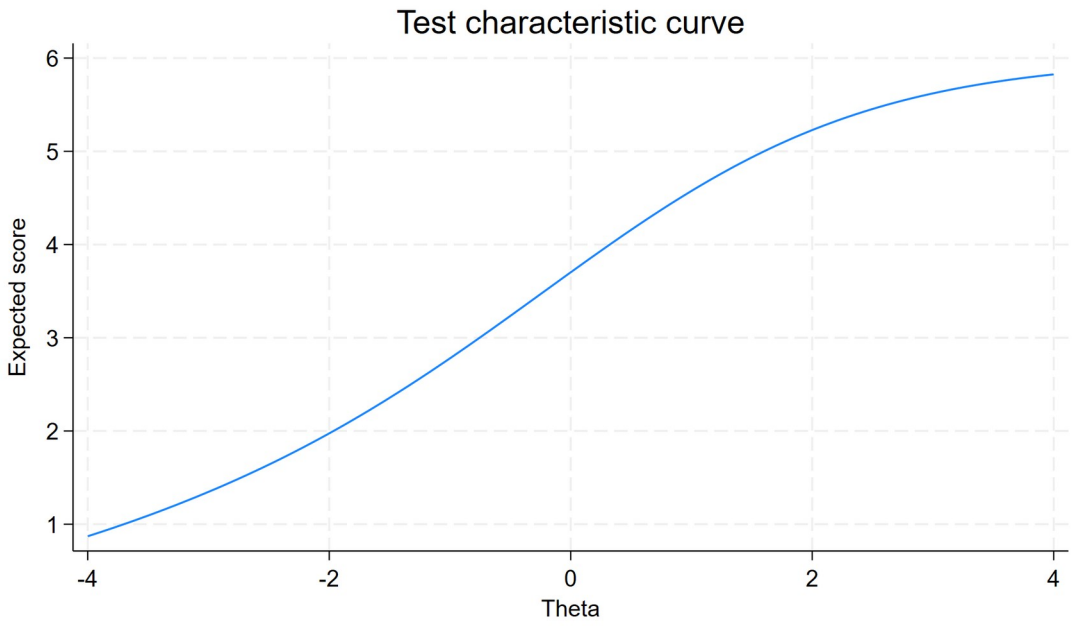
Item 4	79.7%	-1.87	0.403	Easy	Best item-rest relationship; useful item, though somewhat easy.
Item 5	42.4%	0.43	0.107	Moderate-hard	Weak link to the rest of the test; review alignment with construct and distractors/partial-credit logic.
Item 6	50.8%	-0.05	0.026	Near average	Very low item-rest relationship; highest priority for review for ambiguity or construct mismatch.

How to read this table: The strongest item-rest value was Item 4 (0.403), followed by Item 3 (0.306). Item 6 was almost unrelated to the remaining test score, which is a warning sign even though its difficulty level was reasonable.

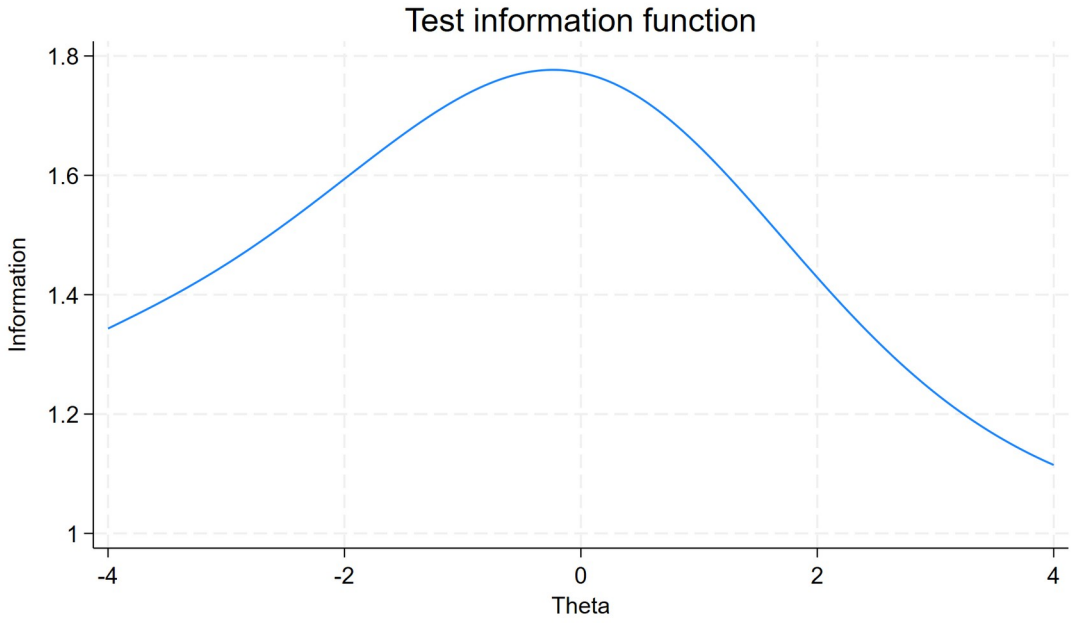
IRT graph interpretation



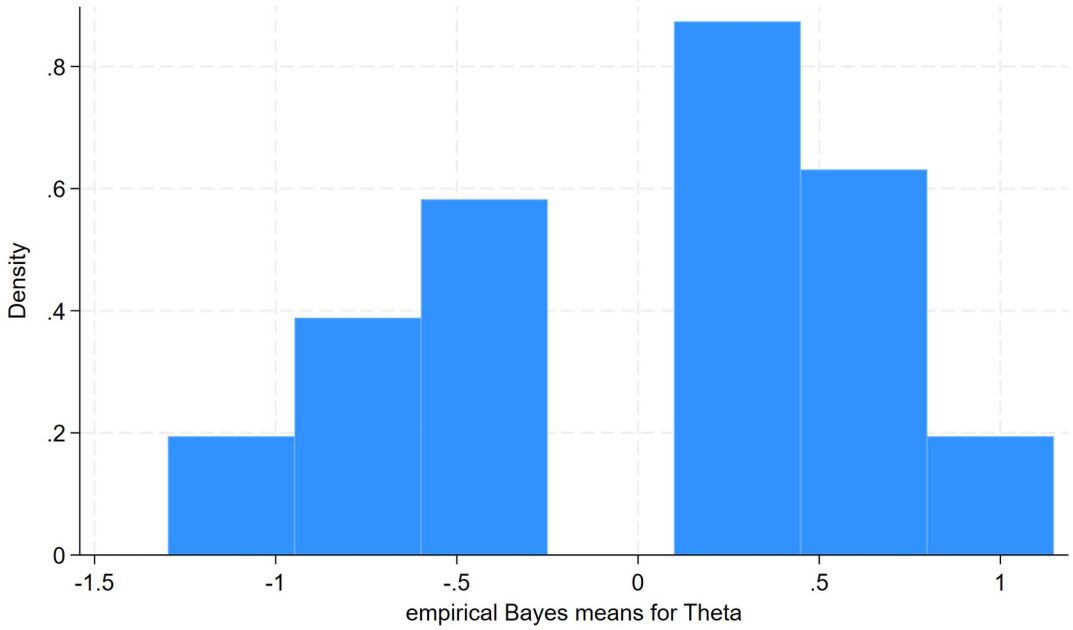
The item characteristic curves confirm the ordering of item difficulty. Item 1 is far to the left and has a high probability of success even for low estimated ability levels. Items 2 and 5 are shifted to the right and are the hardest items. Because this is a Rasch/1PL model, the curves have a common slope; the graph mainly displays differences in item location/difficulty rather than differences in discrimination.



The test characteristic curve is smooth and increasing, as expected: higher estimated ability corresponds to a higher expected score. Around average ability (theta = 0), the expected score is roughly 3.5–4 out of 6. At very high ability, the curve approaches but does not exceed the maximum score of 6.



The test information function peaks near theta = 0 and declines toward both extremes. This means the quiz is most precise for students near the middle of the ability range. It is less informative for identifying fine differences among very low-performing or very high-performing students.



The estimated ability distribution appears split into lower and higher clusters, with relatively few students near theta = 0. With only six items, these theta estimates should be treated as approximate; they are useful for diagnostic interpretation but not for high-stakes individual classification.

Reliability and item relationships

The reliability estimate was alpha = 0.401. For a short six-item quiz, alpha is often lower than for a longer test, but this value still indicates that the items are not strongly working together as a single scale.

Pairwise item correlations were mostly weak. The strongest relationship was between Items 3 and 4 (r = 0.373). Several correlations involving Items 5 and 6 were near zero or negative, including Item 5 with Item 6 (r = -0.049), Item 2 with Item 6 (r = -0.083), and Item 1 with Item 5 (r = -0.029). This supports a content review of Items 5 and 6.

Recommended actions for test authors

1. **1. Review Item 6 first.** It has an appropriate difficulty level but almost no relationship with the rest of the test. Check whether the wording, expected answer, scoring rule, or concept differs from the intended quiz objective.
2. **2. Review Item 5 next.** It is moderately hard but weakly related to the rest of the scale. Check distractors, partial-credit conversion, and whether it measures a secondary skill.
3. **3. Decide the purpose of Item 1.** If it is meant as an introductory confidence item or prerequisite check, it can stay. If it is meant to discriminate among students, it should be made more challenging or replaced.
4. **4. Keep Items 3 and 4 as anchors.** They show the best relationship with total performance and provide useful signal, although Item 4 is on the easier side.
5. **5. Add more items if the quiz is used for decisions.** Six dichotomous items provide limited reliability and measurement precision. Adding targeted items at low, middle, and high difficulty levels would improve the score scale.

6. **6. Re-run the analysis after revisions.** After item edits, repeat the same first-attempt analysis and compare alpha, item-rest correlations, Rasch difficulty estimates, and test information.

Cautions and assumptions

- The report assumes that the binary scoring rules used in the Stata program correctly represent the authors' scoring intentions.
- Because missing item values were summed with rowtotal(), missing item responses may have been treated as incorrect. If missingness reflects non-response or technical issues, the analysis should be rerun with explicit missing-data handling.
- The sample size is adequate for a classroom diagnostic review but small for stable item calibration. Treat exact Rasch difficulty estimates as approximate.
- The Rasch model assumes a common discrimination level for all items. If authors expect items to differ strongly in discrimination, a 2PL model could be explored after more response data are collected.

Appendix: key model outputs

Rasch / 1PL model: N = 59; log likelihood = -194.289; common discrimination = 0.829 (SE = 0.225, $p < .001$); AIC = 402.58; BIC = 417.12. Rasch difficulties: Item 1 = -4.422, Item 2 = 0.722, Item 3 = -0.620, Item 4 = -1.871, Item 5 = 0.428, Item 6 = -0.045.